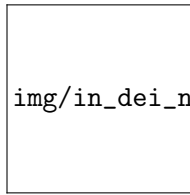


RADBOUD UNIVERSITY



Faculty of Science

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# **Optimizing Local Movement Rules for Heterogeneous Crowd Evacuation with a Genetic Algorithm**

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Natural Computing: NWI-IMC042

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# 1 Introduction

Efficient evacuation is an important problem in settings where many people must move through limited space under time pressure. Crowd-level behaviour emerges from local interactions: individuals move towards an exit, avoid nearby people, and respond to congestion near bottlenecks. These interactions can produce complex collective effects such as clogging, uneven flow, and delayed evacuation.

This project studies a simplified hospital-inspired evacuation scenario with heterogeneous mobility. The simulated crowd contains high-mobility agents, representing staff or visitors, and low-mobility agents, representing patients with reduced walking speed. This heterogeneity is relevant because a movement strategy that improves average evacuation time may still disadvantage slower and more vulnerable individuals.

Our model is inspired by the social force framework introduced by Helbing and Molnár [3], in which pedestrian motion emerges from goal-directed movement and local repulsive forces. Evolutionary methods have previously been used to calibrate continuous pedestrian-model parameters to empirical observations [1] and to optimize evacuation-related conditions [2]. We extend this line of work by using a real-valued genetic algorithm (GA) to optimize interpersonal movement parameters in a heterogeneous crowd model while explicitly considering both evacuation efficiency and mobility-group fairness.

The main research question is:

Can a real-valued genetic algorithm optimize local movement parameters in a heterogeneous crowd model to improve evacuation performance while limiting the evacuation-time gap between high- and low-mobility agents?

We investigate four sub-questions:

1. Does the GA reliably improve the scalar evacuation objective across repeated runs, and which parameter values emerge consistently?
2. How does the fairness weight influence the trade-off between evacuation time, mobility-group disparity, congestion, and near-collisions?
3. How does the allocation of a fixed evaluation budget between population size and number of generations affect optimization performance and robustness?
4. Do evolved local movement parameters remain effective when the proportion of low-mobility agents changes?

## 2 Methods

### 2.1 Evacuation model and scenario

We used a two dimensional agent-based evacuation model implemented in Python, based on a hospital-inspired corridor layout. The environment consists of a rectangular rooms with internal wall segments that form bottlenecks and there is one exit per room on the north- or south-side leading to the corridor, which has two exits located east and west. Agents are circular bodies with a fixed radius and continuous position and velocity. At each discrete time step, all active agents update their velocity and position based on goal-directed acceleration towards an exit, repulsion from nearby agents, and repulsion from walls, after which their speed is capped at a mobility dependent maximum.

The population of the experiment is heterogenous and split into two mobility groups. High mobility agents (maximum speed 1.5 m/s that have a higher maximum speed than low mobility agents (maximum

speed 0.7 m/s, which reflects staff or visitors against fragile patients. In our main `hospital_corridor` experiments, we simulated 40 agents with 28 high-mobility and 12 low-mobility agents (70%/30%). Alternative mixtures were tested in Experiment 4. Agents are placed at random within predefined spawn zones located in the upper and lower rooms, sampled with uniform weights.

Important, each agent is assigned a fixed target exit at initialization based on the shortest path from its spawn location. Agents maintain this route throughout the entire simulation and do not retarget. An evacuation is successful when it reaches within a capture distance of 0.2 m from the room boundary on the exit side and is aligned within the door opening.

Time is discretized into simulation ticks of fixed duration ( $\Delta t = 0.1s$ ). Each run proceeds until either all agents have evacuated or a maximum of 1000 ticks is reached, which is corresponding to 100 s. These limits ensure that incomplete evacuations are penalized while remaining within a plausible time horizon of a single hospital corridor.

## 2.2 Movement parameters and objective

The local movement parameters are controlled by five continuous parameters: the acceleration factor, the agent-agent repulsion weight, the agent-agent interaction radius, the wall repulsion weight, and the wall interaction radius. But in this study the GA only optimizes the first three parameters, which directly govern interpersonal motion. The wall-related parameters are fixed to the layout specific values which are tuned for the hospital scenario.

To evaluate a candidate parameter setting, we ran the simulator and recorder four main performance metrics. Total evacuation time is defined as the simulated time elapsed until the last agent evacuates, or if the maximum evacuation limit is reached and the evacuation is incomplete. Near-collisions count how often any pair of active agents comes within a fixed collision threshold during the run. Mean congestion is calculated as the average number of active agents within a given radius of any exit across all time steps. Fairness is captured by the absolute difference between the mean evacuation times of the high- and low-mobility groups, so that a larger value indicates a stronger disadvantage for one group.

The fitness function used by the genetic algorithm is a weighted sum of these metrics plus a penalty for incomplete evacuations.

$$F = w_{\text{time}} T + w_{\text{coll}} C + w_{\text{cong}} M + w_{\text{fair}} G + P_{\text{incomplete}}.$$

Here,  $T$  is total evacuation time,  $C$  is the total number of near-collisions,  $M$  is the mean congestion near the exits, and  $G$  is the fairness gap between the two mobility groups. The included penalty term  $P_{\text{incomplete}}$  is zero if all agents evacuate within the time limit and otherwise adds a large constant plus an additional cost per remaining agent. For Experiment 1, 3, and 4, we used the default weights  $w_{\text{time}} = 1.0$ ,  $w_{\text{coll}} = 0.05$ ,  $w_{\text{cong}} = 1.0$ , and  $w_{\text{fair}} = 2.0$ . In Experiment 2, we varied the pair  $(w_{\text{time}}, w_{\text{fair}})$  while keeping the collisions and congestions weights fixed, to study the trade-off between throughput and fairness.

## 2.3 Genetic algorithm configuration

we employed a real-valued GA to minimize the fitness function over the three-dimensional space of movement parameters. Each individual encodes a candidate triplet  $[\text{accel\_factor}, \text{agent\_rep\_weight}, \text{agent\_radius}]$  within a predefined lower and upper boundary, Before evaluation, this triplet is expanded into the full parameter vector by appending the fixed wall repulsion parameters.

For each generation, all individuals in the population are evaluated by running the simulator and computing their fitness. We then rank the individuals by fitness and copy the top  $n_{\text{elite}}$  to the next generation.

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Then the remaining positions are filled by offspring generated by tournament selection, arithmetic crossover, and Gaussian mutation with clipping to the allowed parameters boundaries. Tournament selection samples a small subset of individuals and selects the best as parent, which balances exploitation of good solutions with diversity. Arithmetic crossover combines two parents by taking a weighted average of their parameter values. Mutations disturbs each gene independently with a small Gaussian noise with probability  $p_{\text{mut}}$ , scaled relative to the parameter range.

In the default configuration, we used a population size of 24, 25 generations, an elite count of 2, a tournament size of 3, crossover probability 0.9, mutation probability 0.2, and a mutation standard deviation equal to 10% of the parameter range. This yields a budget of 600 candidate evaluations per run. To make the experiment reproducible and robust, a hand-crafted seed solution is injected into the initial population. Every candidate is evaluated on multiple stochastic stimulation seeds, and the metrics are average across those seeds before computing the final fitness, which reduces variance due to random initialization.

## 2.4 Experimental design

All experiments are run on the `hospital_corridor` layout, with the same geometric configuration and population mixture unless stated otherwise. Each GA configuration or fitness weight performed multiple independent GA runs with different random seeds to assess the stability of the optimization process.

### Experiment 1: Baseline convergence and parameter variance.

For the first experiment, we tested the convergence behaviour and parameter variance of the default GA configuration. To do this we ran the GA ten times with different GA random seeds, and keeping the evaluation seeds per candidate fixed. Every generation and run, we recorded the best fitness and corresponding parameter vector. With this we calculated the mean best-fitness trajectory across runs, also we inspected the distribution of the final optimized parameters. To test whether the GA improved consistently upon the starting population, we applied an exact sign test on the initial versus final best-fitness values across seeds. Additionally, we used a Man-Kendall trend test on the mean convergence curve to quantify the monotonicity of fitness improvement over generations.

### Experiment 2: Fitness weight sensitivity and Pareto structure.

For the second experiment, we examined what the influence was of the fitness function and how this influences the trade-offs between evacuation time, fairness, and safety. We defined five distinct  $(w_{\text{time}}, w_{\text{fair}})$  pairs. Those pairs were focused on throughput settings to fairness settings, while keeping the collision and congestion weights fixed to their default values. For each weight, we ran the GA five times with different GA seeds under the default configuration. After we selected the best individual (lowest fitness) per setting for analysis. Those individuals were then re-evaluated on a common set of simulation seeds to obtain robust estimates of mean evacuation time, fairness gap, and near-collisions. Spearman rank correlation was used to quantify the monotonic relationships between the fairness weight and the resulting fairness gap, evacuation time, and near-collisions count.

### Experiment 3: GA hyper parameter ablation.

In the third experiment we investigated how different search dynamics affect optimizations performance under a fixed evaluation budget. We compared three GA configurations, each with 600 candidate evaluations. A default configuration with population size 24 and 25 generations (A), an exploration-based configuration with population 60 and 10 generations (B), and an exploitation-based configuration with population 10 and 60 generations (C). All other parameters were kept identical. Every configuration was run for ten independent

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GA seeds, and we recorder the population mean fitness over evaluations as well as the final best-fitness value per run. The distributions of final best fitness with statistics and applied a Kruskal-Wallis test to detect overall differences between configurations. When significant, we performed pairwise Mann-Whitney tests with Bonferroni correction to identify which configurations differed.

#### **Experiment 4: Population heterogeneity stress test.**

In the last experiment, we evaluated how a single optimized parameter set runs across different mixtures of high- and low-mobility agents. Therefore, we selected the best performing individual from Experiment 1 and kept its movement parameters constant. After that we had three population scenarios in the same hospital layout: the default mixture (20/20), a high mobility-dominant mixture with (32/8) and a low mobility-dominant mixture with (8/32). We ran every scenario for 30 independent simulations with different stochastic seeds and recorded the evaluation time, fairness gap. To check

### **2.5 Implementation and reproducibility**

All simulations and genetic algorithm runs were implemented in a shared codebase. The code contains configuration files for the `hospital_corridor` layout, scripts to run each experiment with the reported hyper parameters and seeds, and the code to reproduce the figures and statistics in Section 3. A README describes the required dependencies, how to set up the environment, and how to execute the main experiments from the command line.

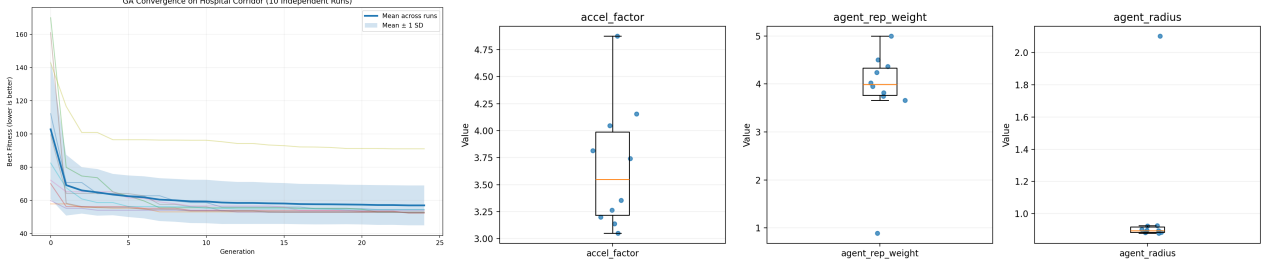
## **3 Results**

### **3.1 Baseline Stability and Parameter Variance**

To evaluate the stability of the GA across stochastic initialisations, we analysed 10 independent runs of the `hospital_corridor` scenario (Experiment 1). The model demonstrated convergence with an exact sign test showing significant improvement in every run (10/10,  $p \approx 0.002$ ). Furthermore, a Mann-Kendall trend test on the mean convergence curve verified strong monotonic progress over the generations ( $S = -298$ ,  $Z = -6.94$ ,  $p < 0.001$ ), as shown in Figure 1.

However, the statistics (mean best fitness:  $56.96 \pm 12.01$  SD) are heavily influenced by a single outlier. Nine of the ten runs converged to a stable state, averaging an evacuation time of 31.0 seconds and approximately 50 near-collisions. On the other hand, run 9 converged to extreme parameter values ( $agent\_radius = 2.10$ ,  $agent\_rep\_weight = 0.89$ ), which resulted in an evacuation time of 28.65 seconds but with 891.7 near-collisions (Figure 1)

When we exclude run 9, the structural variance among the successful runs shows that  $agent\_radius$  converged around 0.90. In contrast,  $agent\_rep\_weight$  had wider variance, ranging from 3.66 to 5.00. The  $accel\_factor$  demonstrated moderate variance, ranging from 3.05 to 4.15 across the successful runs, placing its selection pressure between the strict physical boundaries and the looser repulsion weights.



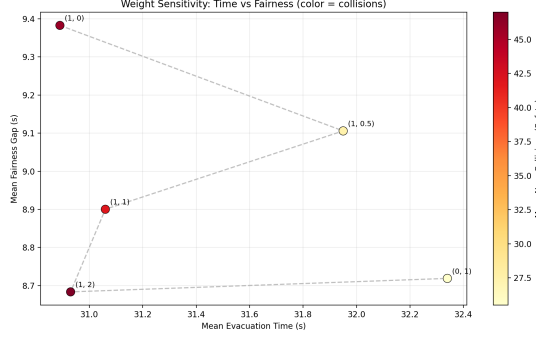
**Figure 1:** Performance and parameter variance across 10 independent GA runs for the `hospital_corridor` scenario. **(Left)** GA convergence over 25 generations. The solid line represents the mean best fitness, with the shaded region indicating  $\pm 1$  SD. The algorithm demonstrates rapid optimization, consistently escaping high initial variance to almost plateau within the first 10 generations. **(Right)** Boxplots that detail the final distribution of the three optimized parameters on independent scales. The distinct spreads reveal differing selection pressures: *agent\_radius* is constrained near 0.90, while *accel\_factor* and *agent\_rep\_weight* demonstrate moderate to high variance. The severe outliers in *agent\_radius* (2.01) and *agent\_rep\_weight* (0.89) belong to run 9, a convergence that fell into a local minimum favouring raw speed over safety.

### 3.2 Weight Sensitivity and Pareto Optimization

To evaluate the multi-objective trade-offs in the `hospital_corridor` scenario, we performed a parameter sweep of the fitness function weights (Experiment 2). We evaluated five distinct weight ratios ( $w_{\text{time}}$  and  $w_{\text{fair}}$ ). To make sure we removed the impact of local minima, each configuration was executed across five independent GA seeds, with the highest fitness candidate selected for analysis. The mean evacuation time, mean fairness gap, and mean near-collisions performance metrics were mapped to approximate an empirical Pareto frontier (Figure 2).

There was a non-linear trade-off between evacuation speed and group fairness in the results. Under a throughput-exclusive objective ( $w_{\text{time}} = 1.0$ ,  $w_{\text{fair}} = 0.0$ ), the system achieved a minimum mean evacuation time of 30.89 seconds, but yielded a high fairness gap of 9.38 seconds and a high collision rate (45.9 near-collisions). Increasing the fairness priority to ( $w_{\text{time}} = 1.0$ ,  $w_{\text{fair}} = 2.0$ ) yielded the absolute minimum fairness gap (8.68 seconds) with only a marginal evacuation delay (30.93 seconds), but this configuration incurred a high safety cost, producing 47.0 near-collisions. Conversely, prioritizing fairness exclusively ( $w_{\text{time}} = 0.0$ ,  $w_{\text{fair}} = 1.0$ ) still achieved a complete evacuation but sacrificed throughput, resulting in the longest evacuation time (32.34 seconds) in exchange for the lowest overall collision count (25.5 near-collisions).

These relationships are supported by non-parametric statistical tests. Spearman rank correlation confirmed a near-perfect negative monotonic relationship between the fairness weight and the resulting fairness gap ( $\rho \approx -0.975$ ). On the other hand, the fairness weight demonstrated weak positive correlations with both evacuation time ( $\rho \approx 0.205$ ) and near-collisions ( $\rho \approx 0.205$ ).



**Figure 2:** Empirical Pareto frontier illustrating the multi-objective trade-off between mean evacuation time and mobility-group fairness gap across five fitness weight ratios ( $w_{\text{time}}$ ,  $w_{\text{fair}}$ ). The colour map encodes the mean near-collisions, serving as a proxy for agent safety. The plot reveals a complex safety cost: while (1, 2) achieves the lowest fairness gap, it incurs high near-collisions (dark red). The optimal safety and fairness balance requires sacrificing evacuation speed, as seen in the (0, 1) configuration (light yellow).

### 3.3 GA Hyper parameter Ablation

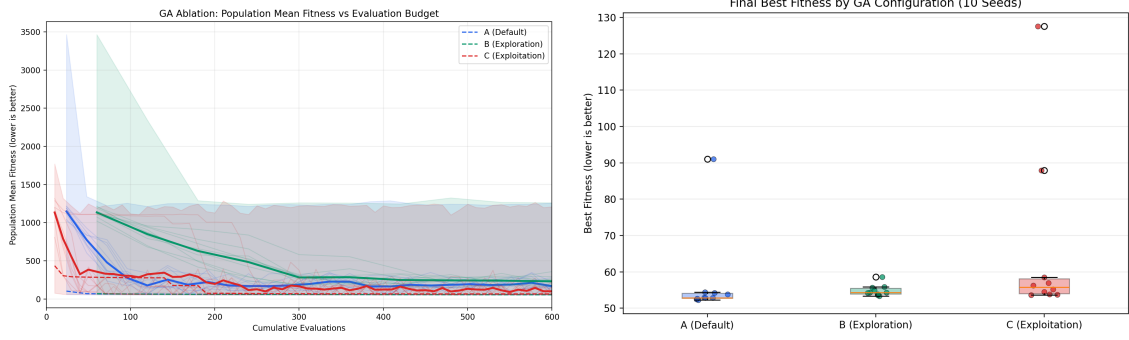
**Table 1:** Summary statistics of final best fitness per GA configuration (10 seeds, lower is better).

| Config           | Median | IQR  | Mean  | Min-Max        |
|------------------|--------|------|-------|----------------|
| A (Default)      | 52.90  | 1.37 | 56.96 | 52.28 – 91.07  |
| B (Exploration)  | 54.33  | 1.56 | 54.88 | 53.38 – 58.58  |
| C (Exploitation) | 55.74  | 4.11 | 65.81 | 53.62 – 127.53 |

We compared three GA configurations on the `hospital_corridor` scenario (Experiment 3) to assess how different search dynamics affect optimisation performance. All configurations used the standard fitness weights, had an evaluation budget of 600 candidate evaluations per run, and were repeated on 10 independent GA seeds (A: population 24, 25 generations; B: population 60, 10 generations; C: population 10, 60 generations). The population mean fitness over the evaluation budget and the distribution of final best fitness per run are shown in Figure 3.

All three configurations showed a rapid decrease in population mean fitness during the first dozen evaluations, followed by a gradual plateau towards the end of the 600 evaluations (Figure 3). The shaded regions demonstrate that configuration C had the largest variability in population mean fitness across seeds. Configuration B showed the narrowest variability band over the full budget.

The final best-fitness distributions varied across the configurations (Table 1). Configuration A (Default) reached a median best fitness of 52.90 with an interquartile range (IQR) of 1.37 and a range from 52.28 to 91.07. Configuration B (Exploration) obtained a median of 54.33 with an IQR of 1.56 and a narrower range, from 53.38 to 58.58. Configuration C (Exploitation) produced a higher median of 55.74, a wider IQR of 4.11, and the widest range, from 53.62 to 127.53. A Kruskal-Wallis test on the final best-fitness values demonstrated that there is a statistically significant difference between at least two configurations ( $H = 7.80$ ,  $df = 2$ ,  $p \approx 0.020$ ). Subsequent post-hoc Mann-Whitney tests with Bonferroni correction showed a significant difference between configurations A and C ( $p_{\text{adj}} \approx 0.047$ ). There was no statistically significant difference after correction between A and B and B and C ( $p_{\text{adj}} \approx 0.103$  and  $0.679$ , respectively).



**Figure 3: Left:** Convergence of population mean fitness over 600 evaluations for three GA configurations. The shaded regions indicate variance across 10 independent seeds. Configuration C (exploitation) exhibits a rapid initial descent but quickly plateaus, leading to higher instability in later stages. Configuration A (default) has the most stable and optimal convergence. **Right:** Distribution of final best fitness values across 10 different seeds. Lower fitness means better performance. Configuration A (default) is the most robust, it achieved the lowest median fitness with minimal variance. Conversely, Configuration C (exploitation) produces high-fitness outliers, confirming that over-exploitation traps the algorithm in local optima.

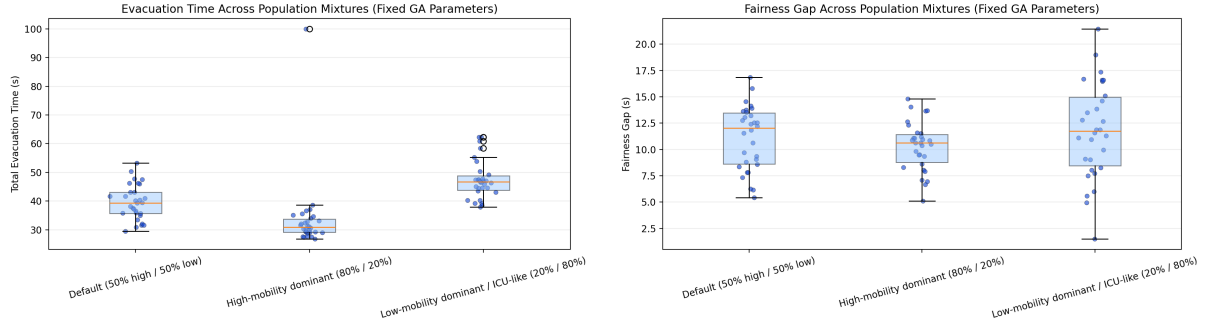
### 3.4 Population Heterogeneity Stress Test

We re-evaluated a fixed GA parameter set (from Experiment 1, seed 606), to assess how the optimised movement parameters generalise across different populations. We had three population mixtures in the `hospital_corridor` layout: Default (20 high / 20 low), High-mobility (32 high / 8 low), and low mobility dominant ICU-like (8 high / 32 low). Each scenario was evaluated over 30 independent simulation seeds, and all runs completed evacuation except for a single high mobility dominant run, resulting in completion rates of 100%, 96.7%, and 100% respectively.

Evacuation times differed between the scenarios (Figure 4). The Default mixture yielded a median total evacuation time of 39.35 s (IQR 7.38 s, range 29.5-53.3 s), while the high mobility dominant achieved a lower median of 30.90 s (IQR 4.55 s, range 26.8-100.0 s). The low mobility mixture had the slowest evacuation time, with a median of 46.65 s (IQR 5.03 s, range 37.9-62.3 s). The Kruskal-Wallis test on total evacuation time demonstrated that there is a significant difference across the three populations ( $H = 50.48$ ,  $df = 2$ ,  $p \approx 1.09 \times 10^{-11}$ ).

On the other hand, fairness gaps remained closer across the scenarios (Figure 4). The default and low mobility mixtures had median fairness gaps of 12.02 s and 11.73 s, with IQRs of 4.82 s and 6.51 s. The high mobility group demonstrated a lower median fairness gap of 10.62 s and the smallest IQR (2.64 s). However, the Kruskal-Wallis test on fairness gaps did not detect any statistically significant differences between the groups ( $H = 2.84$ ,  $df = 2$ ,  $p \approx 0.24$ ).





**Figure 4: Left:** Total evacuation time across different populations using fixed GA parameters. The high mobility mixture predictably achieves the lowest median evacuation time; however, it has some outliers that did not evacuate, suggesting that spatial bottlenecks can still cause critical failures. **Right:** Fairness gap across the same mixtures. The low mobility group shows the highest variance and largest quartile fairness gap, indicating that GA struggles to consistently balance evacuation equity with slow agents outnumbering fast agents.

## 4 Discussion

The experiments show that a real-valued GA can optimize local movement parameters in the simplified heterogeneous evacuation model. Across ten independent runs, the GA improved the scalar objective consistently and typically approached a plateau within the first ten generations. This indicates that evolutionary search can identify useful parameter regions efficiently under the chosen evaluation budget. However, the parameter distributions also show that multiple solutions can produce similar fitness values. The optimized acceleration factor and repulsion weight varied more strongly than the interaction radius, suggesting that the objective constrains some aspects of the movement dynamics more strongly than others.

The outlier in Experiment 1 is particularly informative. This run achieved a relatively short evacuation time but accumulated a very large number of near-collisions. The result exposes a limitation of the scalar objective: although near-collisions are penalized, the collision term does not impose a hard safety constraint. A fast but undesirable solution can therefore remain competitive if improvements in other fitness components compensate for a poor safety outcome. This suggests that future work should investigate stronger collision penalties, explicit feasibility constraints, or a multi-objective evolutionary algorithm that preserves separate efficiency and safety objectives.

Experiment 2 demonstrates that the fitness coefficients materially influence the behaviour that evolves. Increasing the fairness weight generally reduced the evacuation-time gap between mobility groups, but it did not automatically improve evacuation time or near-collision counts. The objective weights should therefore not be treated as neutral technical parameters. They encode priorities between throughput, fairness, congestion, and safety. In practical applications, these priorities would require careful justification by domain experts.

The GA hyperparameter ablation in Experiment 3 shows that the allocation of a fixed evaluation budget affects both performance and robustness. Configuration A achieved the lowest median final fitness, while Configuration B produced a narrower range of outcomes across seeds. Configuration C showed the largest variability and the most severe high-fitness outliers. These results suggest that using too small a population can make the search more prone to poor outcomes, even when the total number of candidate evaluations is unchanged. However, the current experiment does not establish the underlying optimization mechanism conclusively.

The population heterogeneity stress test shows that the fixed optimized parameters remain usable when the population composition changes. As expected, the low-mobility-dominant scenario required more time to evacuate than the high-mobility-dominant scenario. Differences in fairness gap were not statistically

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significant. The optimized local rules therefore do not remove the throughput cost of a more vulnerable population, but they remain functional across the tested mixtures.

Several limitations should be considered. First, the hospital-inspired corridor is a simplified environment rather than an empirically validated hospital floor plan. Second, the local motion rules are simplified and are not calibrated against pedestrian-trajectory data. Third, near-collisions are a proxy for safety rather than a validated safety measure. Fourth, the generalization experiment varies mobility-group ratios but does not test unseen layouts, crowd densities, or exit widths. Finally, the use of a weighted scalar objective compresses several competing goals into a single fitness value and can hide important trade-offs.

Future work should evaluate additional layouts, crowd densities, and bottleneck widths; compare the GA against fixed heuristics and random search under the same hospital scenario; calibrate movement parameters against empirical pedestrian data; and investigate true multi-objective evolutionary optimization. Hard safety constraints could also prevent fast but unsafe solutions from being selected.

## 5 Conclusion

A real-valued GA can improve the scalar evacuation objective in this simplified heterogeneous crowd model. The evolved behaviour depends strongly on the selected fitness weights and GA configuration, and optimization does not automatically guarantee safe outcomes. The optimized parameters remained functional across different mobility-group ratios, although low-mobility-dominant populations still required more evacuation time.

The study therefore supports evolutionary optimization as a useful method for investigating emergent crowd dynamics in a controlled simulation setting. Further validation and more extensive generalization experiments are required before the model can support practical evacuation-planning decisions.

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